EXERCISE 11

Euler-Cauchy Equation via Exponential Transformation

Original statement · ProblemsCh3.pdf, p. 3

Euler-Cauchy • Coordinate Mapping • Resonance

PHASE 1: STATEMENT & SETUP

We consider the following second order ODE (Euler equation):

$$x^2y''(x) + pxy'(x) + qy(x) = g(x), \quad x > 0$$

1. By making the change of variable $x = e^z$ and using the chain rule, show that the ODE is transformed into one with constant coefficients in the variable z:

$$\frac{d^2y}{dz^2} + (p-1)\frac{dy}{dz} + qy = g(e^z)$$

2. Take $p = 3,\ q = -3$ and $g(x) = 4x$, make the previous transformation and compute the general solution.

ERRATUM IN THE PRINTED SHEET

In ProblemsCh3.pdf, Exercise 11 (i), the transformed equation is printed as $\frac{d^2y}{dz^2} + (p-1)\frac{dy}{dz} + y = g(e^z)$, i.e. with "$+\,y$". The correct coefficient is q (the term $q\,y(x)$ of the original equation is unchanged by $x = e^z$), so the equation above, with "$+\,q\,y$", is the one derived in Phase 3 and used in part 2.

PHASE 2: FUNDAMENTAL FORMULATION

Let $x = e^z \iff z = \ln x$ for $x > 0$. The derivative with respect to x is $\frac{dz}{dx} = \frac{1}{x}$. By applying the chain and product rules to $\frac{dy}{dx}$ and $\frac{d^2y}{dx^2}$, the power pre-factors x and x^2 cancel identically, transforming the variable-coefficient ODE into a linear ODE with constant coefficients.

METHOD OF THE CLASS NOTES [NOTES, PP. 5-6]

Part (ii) reduces to $\frac{d^2y}{dz^2} + 2\frac{dy}{dz} - 3y = 4e^z$, solved as in the notes: $y = y_H + y_N$. $k^2 + 2k - 3 = (k+3)(k-1)$ gives Case 01, $y_H = C_1 e^z + C_2 e^{-3z}$. The forcing $4e^z$ has $\alpha = 1$, a root, so the educated guess is $y_N = Aze^z$ (Case 2 of the guess).

With $(Aze^z)' = A(1+z)e^z$ and $(Aze^z)'' = A(2+z)e^z$: $A\big[(2+z) + 2(1+z) - 3z\big]e^z = 4Ae^z = 4e^z$, so $A = 1$ and, with $z = \ln x$, $y = C_1 x + C_2 x^{-3} + x\ln x$.

PHASE 3: STEP-BY-STEP ANALYTICAL DERIVATION

Part (i): Chain Rule Derivation of the Transformation

First Derivative:

$$\frac{dy}{dx} = \frac{dy}{dz}\frac{dz}{dx} = \frac{1}{x}\frac{dy}{dz} \implies x\frac{dy}{dx} = \frac{dy}{dz} \tag{1}$$

Second Derivative:

$$\frac{d^2y}{dx^2} = \frac{d}{dx}\left(\frac{1}{x}\frac{dy}{dz}\right) = -\frac{1}{x^2}\frac{dy}{dz} + \frac{1}{x}\frac{d}{dx}\left(\frac{dy}{dz}\right) = -\frac{1}{x^2}\frac{dy}{dz} + \frac{1}{x}\left(\frac{d^2y}{dz^2}\frac{dz}{dx}\right)$$

$$\frac{d^2y}{dx^2} = -\frac{1}{x^2}\frac{dy}{dz} + \frac{1}{x^2}\frac{d^2y}{dz^2} = \frac{1}{x^2}\left(\frac{d^2y}{dz^2} - \frac{dy}{dz}\right) \implies x^2\frac{d^2y}{dx^2} = \frac{d^2y}{dz^2} - \frac{dy}{dz} \tag{2}$$

Substitution into the Euler-Cauchy Equation:

$$x^2y'' + pxy' + qy = \left(\frac{d^2y}{dz^2} - \frac{dy}{dz}\right) + p\left(\frac{dy}{dz}\right) + qy = g(e^z)$$

$$\mathbf{\frac{d^2y}{dz^2} + (p-1)\frac{dy}{dz} + qy = g(e^z)}$$

`Part (ii): Specific Case` $p = 3,\ q = -3,\ g(x) = 4x$

1. Transformed ODE in z:

With $p - 1 = 3 - 1 = 2, q = -3$, and $g(e^z) = 4e^z$:

$$\frac{d^2y}{dz^2} + 2\frac{dy}{dz} - 3y = 4e^z$$

2. Homogeneous Solution $y_h(z)$:

Auxiliary eq: $r^2 + 2r - 3 = 0 \implies (r+3)(r-1) = 0 \implies r_1 = 1, r_2 = -3.$

$$y_h(z) = c_1e^z + c_2e^{-3z}$$

3. Resonant Particular Solution $y_p(z)$:

Since the forcing exponent $e^{1\cdot z}$ matches the characteristic root $r_1 = 1$, we test the resonant ansatz $y_p(z) = Aze^z$:

$$y_p'(z) = A(1+z)e^z, \qquad y_p''(z) = A(2+z)e^z$$

$$y_p'' + 2y_p' - 3y_p = Ae^z\left[(2+z) + 2(1+z) - 3z\right] = Ae^z\left[4 + 3z - 3z\right] = 4Ae^z$$

Equating to $4e^z$: $4A = 4 \implies A = 1 \implies \mathbf{y_p(z) = ze^z}$.

4. General Solution in z:

$$y(z) = c_1e^z + c_2e^{-3z} + ze^z$$

5. Back-transformation to x ($z = \ln x,\ e^z = x,\ e^{-3z} = x^{-3}$):

$$\mathbf{y(x) = c_1 x + \frac{c_2}{x^3} + x \ln x, \quad x > 0}$$

PHASE 4: MASTER VERIFICATION & AEROSPACE APPLICATION

MASTER SOLUTION & ODE VERIFICATION

$$\boxed{y(x) = c_1 x + c_2 x^{-3} + x \ln x, \quad x > 0}$$

Direct Verification:

$y'(x) = c_1 - 3c_2 x^{-4} + \ln x + 1$

$y''(x) = 12c_2 x^{-5} + \frac{1}{x}$

$x^2 y'' + 3xy' - 3y = (12c_2 x^{-3} + x) + (3c_1 x - 9c_2 x^{-3} + 3x \ln x + 3x) - (3c_1 x + 3c_2 x^{-3} + 3x \ln x) = 4x \equiv g(x)$. Confirmed!